**Batch: DMW 2 Roll No.: 16010324044**
**Experiment/Assignment/Tutorial No.: 01**
**Grade: AA / AB / BB / BC / CC / CD /DD**
**Signature of the Staff In-charge with date**

**TITLE: Association rule mining**

**AIM:** To implement the Apriori algorithm and validate its correctness and efficiency using the datasets provided.

**OUTCOME:** By the end of this experiment, students will:

1. Perform EDA to learn data characteristics
2. Implement Apriori algorithm:
   - To extract association rules from the dataset
   - To analyse effect of minsupport and minconfidence on rule extraction and time duration
3. Use Python libraries like pandas, numpy, matplotlib and **mlxtend** to preprocessing, visualization and association rule mining **.**

**Procedure:**

1. **Load the dataset**:
   - o Import data from book.csv file
2. **Perform data pre-processing**:
   - o Look for missing value, data formatting requirements
3. **Perform Exploratory data analysis**:
   - o Use appropriate descriptive statistics and graphs to report data characteristics.
4. **Apply Apriori algorithm to extract association rules :**(pip install **mlxtend)**
   - o Make a table showing for each level (i.e., for each size of the itemsets) of the mining process (a) how many candidates are generated by your code and (b) how many of them are frequent. For example, your table may look like this:
   - o You may consider minimum support = 0.25

| Level (Number of items) | 1 | 2 | 3 | 4 | .. |
|---|---|---|---|---|---|
| Candidate itemsets | e.g. 563 | | | | |
| Frequent itemsets | e.g. 135 | | | | |

5. **Prepare a similar table for Association rules as in 4 above**
6. **Design and output of a Recommendation system for a single and multiple input items**
7. **Repeat above procedure for “store_data.csv” and report your observations and results as a brief summary.**
8. **Explore Weka tool for association rule mining and note your observations**

**Observation Table:**

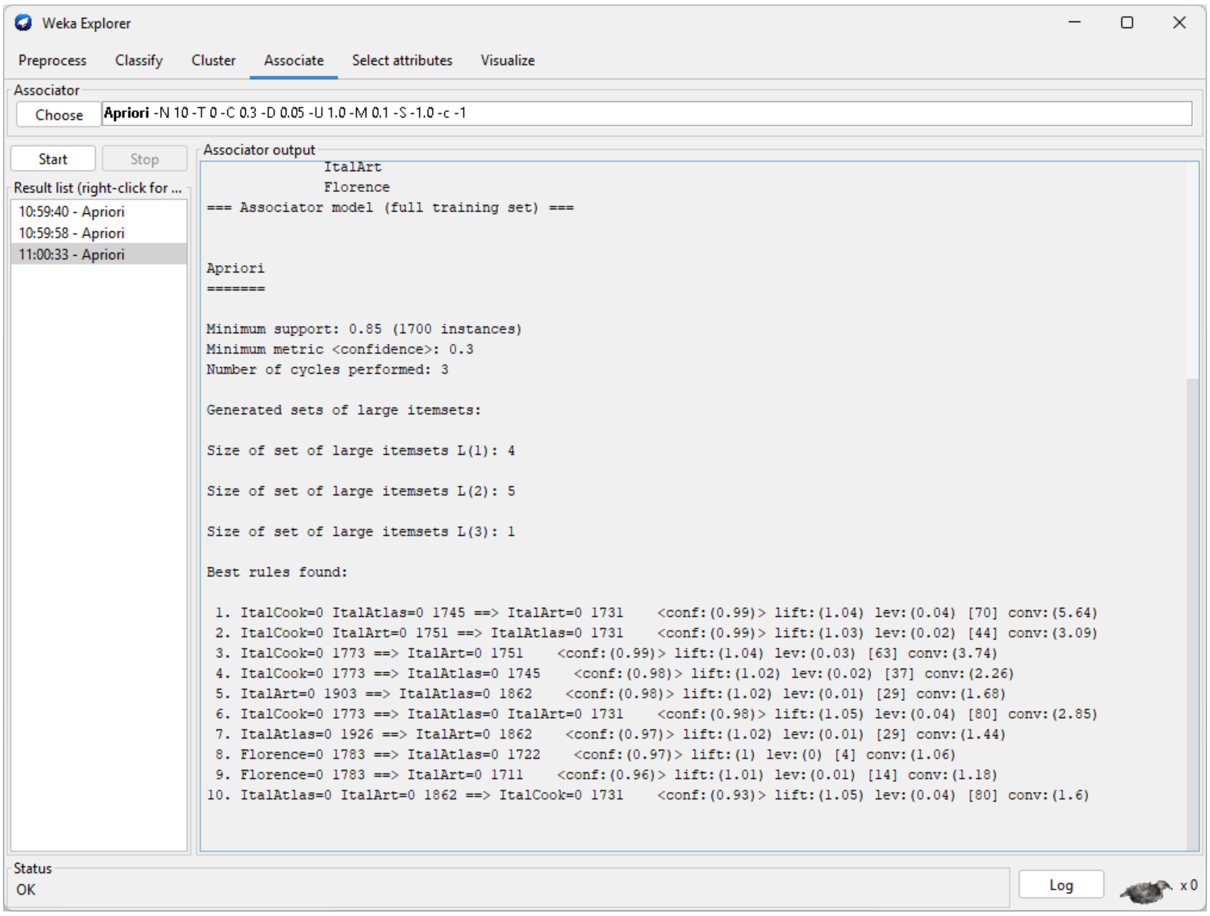
| **Procedure Step** | **Data set: book.csv** | |
|---|---|---|
| | **Observation\Output** | **Interpretation\remark** |
| **Step 1** | Generate Frequent Itemsets<br>How many frequent itemsets are generated using minimum support = 0.10?<br>39 | Frequent itemsets are groups of books purchased together at least 10% of the time. |
| **Step 2** | Which frequent itemset has the highest support? Write its support value.<br>Frequent Itemset Support<br>0.4310, CookBks | This is the most frequently occurring book/itemset in the dataset. |
| **Step 3** | List any three frequent itemsets generated by the Apriori algorithm.<br>Sr. No. Frequent Itemset Support<br>0 0.4230 (ChildBks)<br>1 0.2475 (YouthBks)<br>2 0.4310 (CookBks) | These itemsets satisfy the minimum support threshold and represent common purchase patterns. |
| **Step 4** | Generate Association Rules<br>How many association rules are generated using minimum confidence = 0.60?<br>30 | These rules indicate strong relationships where the consequent is likely to occur when the antecedent occurs. |
| **Step 5** | Write any three association rules generated.<br>Antecedent Consequent<br>11 (ItalCook) (CookBks)<br>23 (ChildBks, ArtBks) (GeogBks) | These rules reveal purchasing patterns useful for recommendations. |

| | | |
|---|---|---|
| | 28 (ArtBks, CookBks) (GeogBks) | |
| **Step 6** | Select the association rule with the highest confidence and interpret it.<br>Rule Confidence<br>(ItalCook) (CookBks) 1.000000 | This rule has the highest probability of occurring and is the most reliable among all generated rules. |
| **Step 7** | Select the association rule with the highest lift.<br>Rule Lift<br>(ItalCook) (CookBks) 2.320186 | This rule represents the strongest positive association between items because its lift value is maximum. |
| **Step 8** | Explain the meaning of the following values obtained for any one rule.<br>Support = frequency of combination<br>Confidence = reliability of combination<br>Lift = strength of combination | Support measures frequency, Confidence measures reliability, and Lift measures the strength of association compared to random chance. |
| **Step 9** | Sort the rules in descending order of Lift and list the Top 5 Rules.<br>Rank Rule Lift<br>11 (ItalCook) (CookBks) 0.1135 1.000000 2.320186<br>23 (ChildBks, ArtBks) (GeogBks) 0.1020 0.627692 2.274247<br>28 (ArtBks, CookBks) (GeogBks) 0.1035 0.619760 2.245509<br>26 (ArtBks, CookBks) (DoItYBks) 0.1015 0.607784 2.155264<br>25 (DoItYBks, ArtBks) (CookBks) 0.1015 0.821862 1.906873 | Rules with higher lift indicate stronger associations and are more useful for recommendation systems. |
| **Step 10** | Based on the generated association rules, write three recommendations for an online bookstore.<br>Customer Purchased = ItalCook<br>Recommended Item = Discount or Bundling on CookBks based on lift | Products are recommended based on previously discovered buying patterns from the association rules. |

<table>
<tr><th>Procedure Step</th><th colspan="2">Data set: store_data.csv</th></tr>
<tr><td></td><td>Observation\Output</td><td>Interpretation\remark</td></tr>
<tr><td>Step 1</td><td>How many frequent itemsets are generated using minimum support = 0.02?<br>103</td><td>Lower support allows more frequent itemsets to be discovered because even less frequent combinations are included.</td></tr>
<tr><td>Step 2</td><td>Write the Top 5 frequent itemsets obtained.<br>Frequent Itemset Support<br>8 (ground beef) (spaghetti) 0.039195 0.398915 2.291162<br>18 (olive oil) (spaghetti) 0.022930 0.348178 1.999758<br>14 (soup) (mineral water) 0.023064 0.456464 1.914955<br>0 (burgers) (eggs) 0.028796 0.330275 1.837830<br>11 (olive oil) (mineral water) 0.027596 0.419028 1.757904</td><td>These are the most commonly purchased item combinations in supermarket transactions.</td></tr>
<tr><td>Step 3</td><td>How many association rules are generated using minimum confidence = 0.30?<br>20</td><td>Lower confidence generates more association rules, including weaker relationships.</td></tr>
<tr><td>Step 4</td><td>Write any three strong association rules.<br>Antecedent Consequent<br>(ground beef) (spaghetti)<br>(olive oil) (spaghetti)<br>(soup) (mineral water)</td><td>These rules indicate products that are frequently purchased together.</td></tr>
<tr><td>Step 5</td><td>Identify the rule having the highest confidence and interpret</td><td>Customers purchasing the antecedent are most likely to</td></tr>
</table>

| | | |
|---|---|---|
| | Rule Confidence<br>14 (soup) (mineral water) 0.023064 0.456464 1.914955 | purchase the consequent. This rule has the highest predictive accuracy. |
| **Step 6** | Identify the rule having the highest lift **and interpret**<br>Rule Lift<br>8 (ground beef) (spaghetti) 0.039195 0.398915 2.291162 | This rule has the strongest association beyond random chance and is highly valuable for recommendations. |
| **Step 7** | Interpret the following for any one selected rule.<br>Support = frequency of combination<br>Confidence = reliability of combination<br>Lift = strength of combination | Support indicates frequency, Confidence indicates reliability, and Lift indicates the strength of association between products. |
| | Based on the generated rules, write three product recommendations for a supermarket.<br>Purchased Item<br>olive oil<br>Recommended Item<br>discount and bundling on spaghetti, mineral water based on lift | The recommendations help supermarkets increase cross-selling by suggesting products commonly bought together. |

## Weka tool:




## Observation:

If ItalCook is zero and ItalAtlas is also zero, then with 99% confidence we can say that ItalArt will also be zero with 1.04 times more likely to occur and 4% more co-occurrence than expected.

## Conclusion:

In this experiment, I learnt about the Apriori algorithm and successfully applied to discover frequent itemsets and association rules from the datasets. The experiment demonstrated how support, confidence, and lift are used to identify meaningful relationships among items. The generated rules can be effectively used in recommendation systems for applications such as online bookstores and supermarkets.

**Signature of faculty in-charge with date**